



CS1 exercise session

About the exercise

- ▷ Slides from Joel Gmür
- ▷ Intuitive understanding
- ▷ Visualisations will be added
- ▷ All content on website

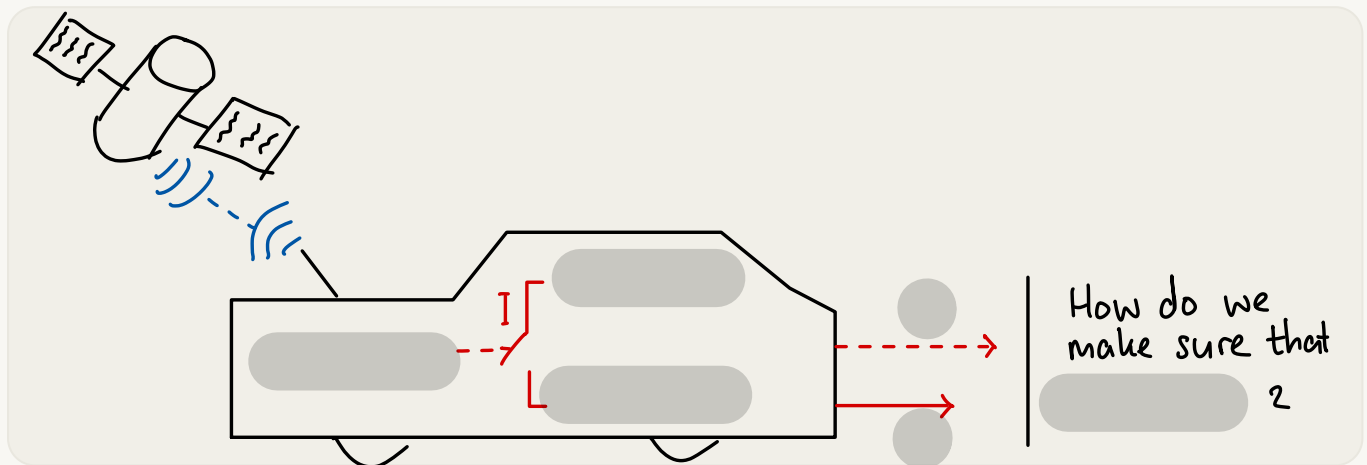
About the lecture

- ▷ 2-page summary sheet given — 40 pages handwritten summary on top
- ▷ Lecture notes PDF
- ▷ Study Center starts 3rd week



What is control?

Let's build intuition with an example.



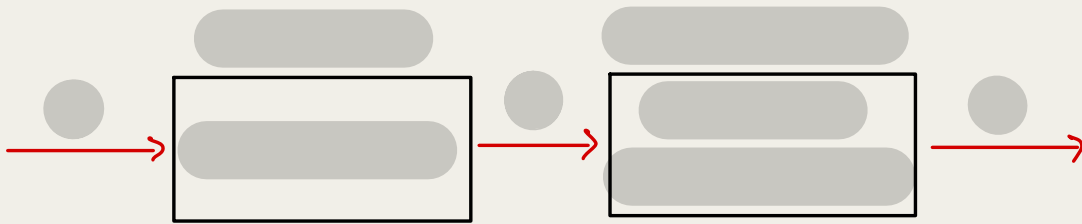
We command the car's computer to move the car with _____. But the computer can only control the _____ going into the motor — it knows the car's true speed through _____ of the satellite.

How does the computer know how much current it should send?

— *this is the central question of control.*

To solve this we will first need to learn many techniques. For you to keep an overview, you must keep the greater picture in mind.

So let's abstract the car example (*for now without the satellite*).



In the sketch, V_0 , I , V_m are called _____, while the computer and the motor are called _____.

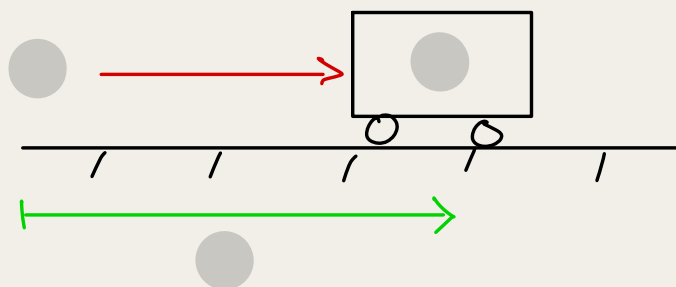
A system _____ a signal.

What are signals and systems?

Signal — a _____ of _____ (or _____).

System — any _____ that possibly has _____ and/or _____.

Example:



mass m
force $F(t)$
position x

System — the _____ with mass m

Signals — _____,

We can model systems with differential equations.

Here:

$$\dot{X}(t) = \text{_____}$$

$$X(t) = \text{_____}$$

From initial conditions: $C_1 = \text{_____}$, $C_2 = \text{_____}$

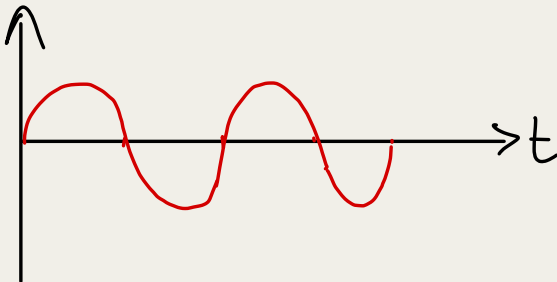
$$X(t) = \text{_____}$$

Assume

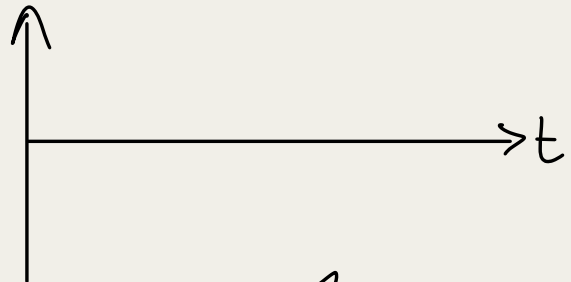
$$X_0 = \text{_____}, V_0 = \text{_____}$$

In the end, the system is the underlying equation _____ and the _____. It defines how the input signal is manipulated.

$$F(t) = \sin(t)$$

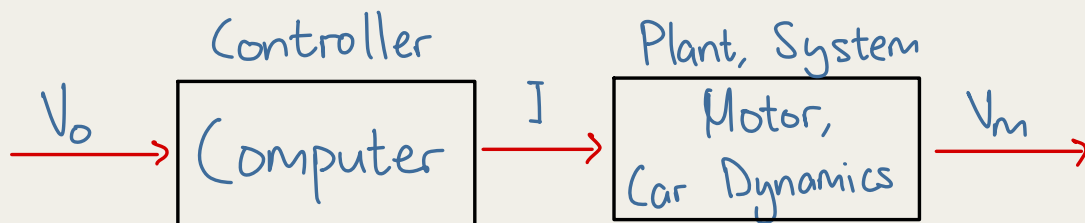
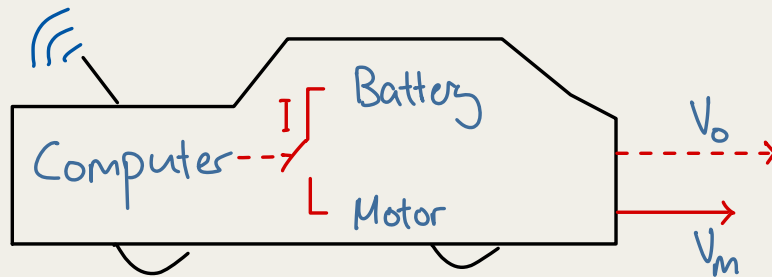


$$x(t) = ?$$



System

Wait, why did we just look at this?

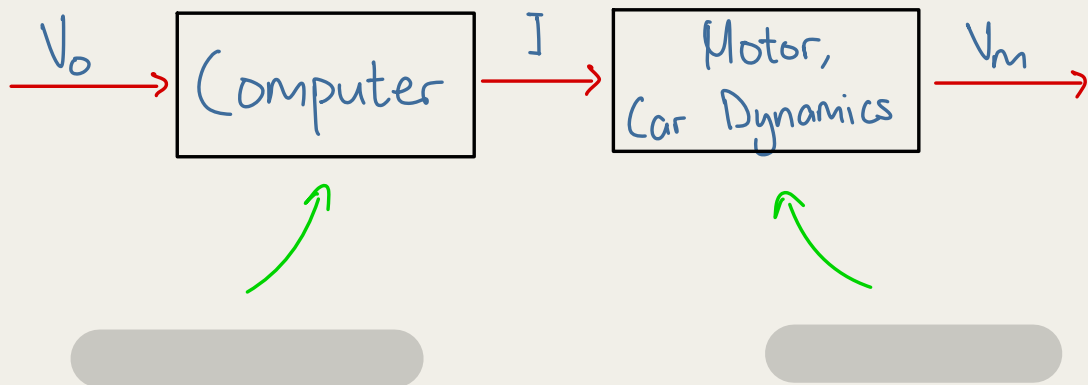


 Blocks are systems → Arrows are signals

To figure out how to _____ the car, we need to know the diff' eqns that describe the behaviour of the blocks — this is why the first part of the course will be _____.

If every system is just defined by physics,
then what even is our job as engineers?

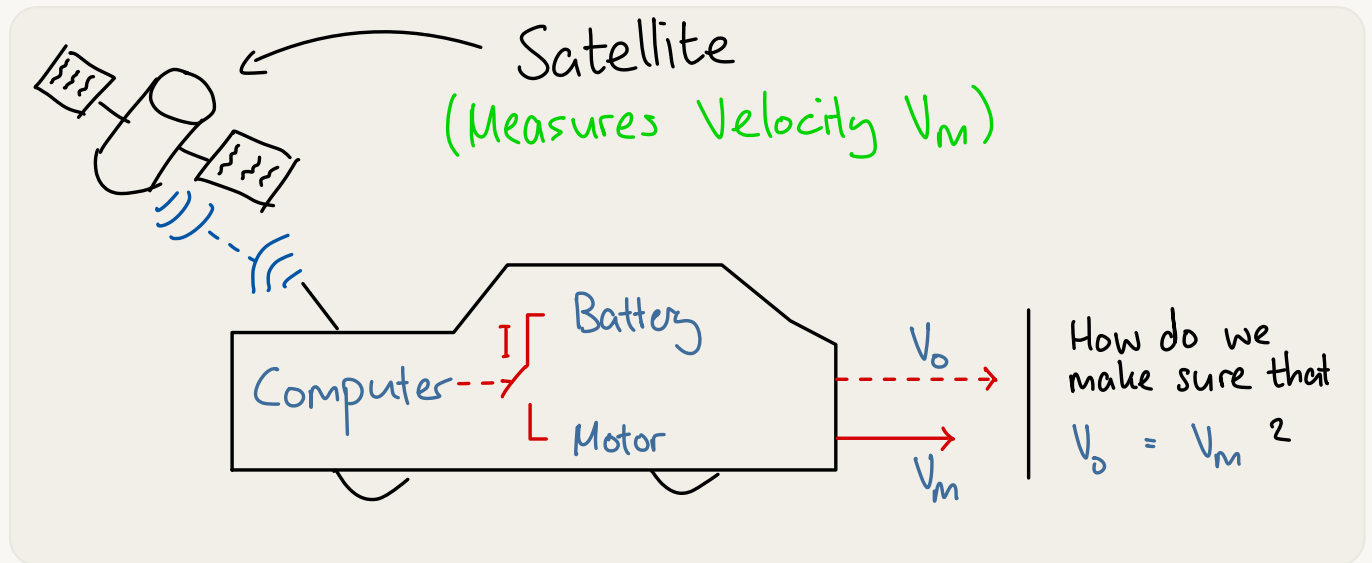
— *this is not true!!*



Some blocks are predefined. The ones that we can design (computer), are called _____. We need to design controllers such that the _____ does what we want. This is hard because of:

1. _____
2. _____
3. _____
- ...

These problems will be addressed by introducing Back to our example:



Think about how the satellite could improve this problem...

All of what I showed you was massively simplified and not covered yet. (you are not expected to understand it now) — this was just so you have the big picture in mind.

Intermediate Quiz

Q1) The Input is what we tell the computer to do. True / False

Q2) Each block represents a physical system. True / False

Q3) Not all control systems are electrical. True / False